INVERTIS UNIVERSITY BAREILLY

Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

BCS-091	Cryptography and Network Security	L T P 3 0 0	3 Credits

Pre-requisites: Graduate student status or a Senior in Computer Science

Course Objectives:

CO1	To introduce fundamental concepts of symmetric and asymmetric cipher models
CO2	To introduce fundamental concepts of authentication.
CO3	To introduce network security and web security protocols.
CO4	To understand DES, AES
CO5	To understand Discrete Logarithmic Problem

Detailed Syllabus

MODULE-I

Introduction to security attacks: services and mechanism, Classical encryption techniques substitution ciphers and transposition ciphers, cryptanalysis, steganography, Stream and block ciphers.

Modern Block Ciphers: Block ciphers principles, Shannon's theory of confusion and diffusion, fiestal structure, Data encryption standard (DES), Strength of DES, block cipher modes of operations, Triple DES, Advanced Encryption Standard (AES) encryption and decryption.

MODULE-II

Introduction to group, field, finite field of the form GF(p): modular arithmetic, prime and relative prime numbers, Extended Euclidean Algorithm, Fermat's and Euler's theorem, Primality testing, Chinese Remainder theorem, Discrete Logarithmic Problem, Principals of public key crypto systems, RSA algorithm, security of RSA, Message Authentication Codes: Authentication requirements, authentication functions, message authentication code, hash functions, birthday attacks, security of hash functions, Secure hash algorithm(SHA), **Digital Signatures**: Digital Signatures, Elgamal Digital Signature Techniques, Digital Signature Standards (DSS).

MODULE-III

Key Management and distribution: Symmetric key distribution, Diffie-Hellman Key Exchange, Public key distribution, X.509 Certificates, Public key Infrastructure, Authentication Applications: Kerberos Electronic mail security: pretty good privacy (PGP), S/MIME, IP Security: Architecture, Authentication header, Encapsulating security payloads, combining security associations, key management. Introduction to Secure Socket Layer, Secure electronic, transaction (SET), System Security: Introductory idea of Intrusion, Intrusion detection, Viruses and related threats, firewalls.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Analyze solutions for effective key management and distribution and conduct cryptanalysis
CO2	Analyze and use cryptographic data integrity algorithms and user authentication protocols
CO3	Analyze the security requirements and solutions for wireless networks and distributed systems
CO4	Analyze Discrete Logarithmic Problem
CO5	Explore the attacks and controls associated with IP, transport-level, web and E-mail security

Text Books:

1. William Stallings, “Cryptography and Network Security: Principals and Practice”,Pearson Education.
2. Behrouz A. Frouzan: Cryptography and Network Security, Tata McGraw Hill

References:

1. Bruce Schiener, “Applied Cryptography”. John Wiley & Sons
2. Bernard Menezes,” Network Security and Cryptography”, Cengage Learning.
3. AtulKahate, “Cryptography and Network Security”, Tata McGraw Hill